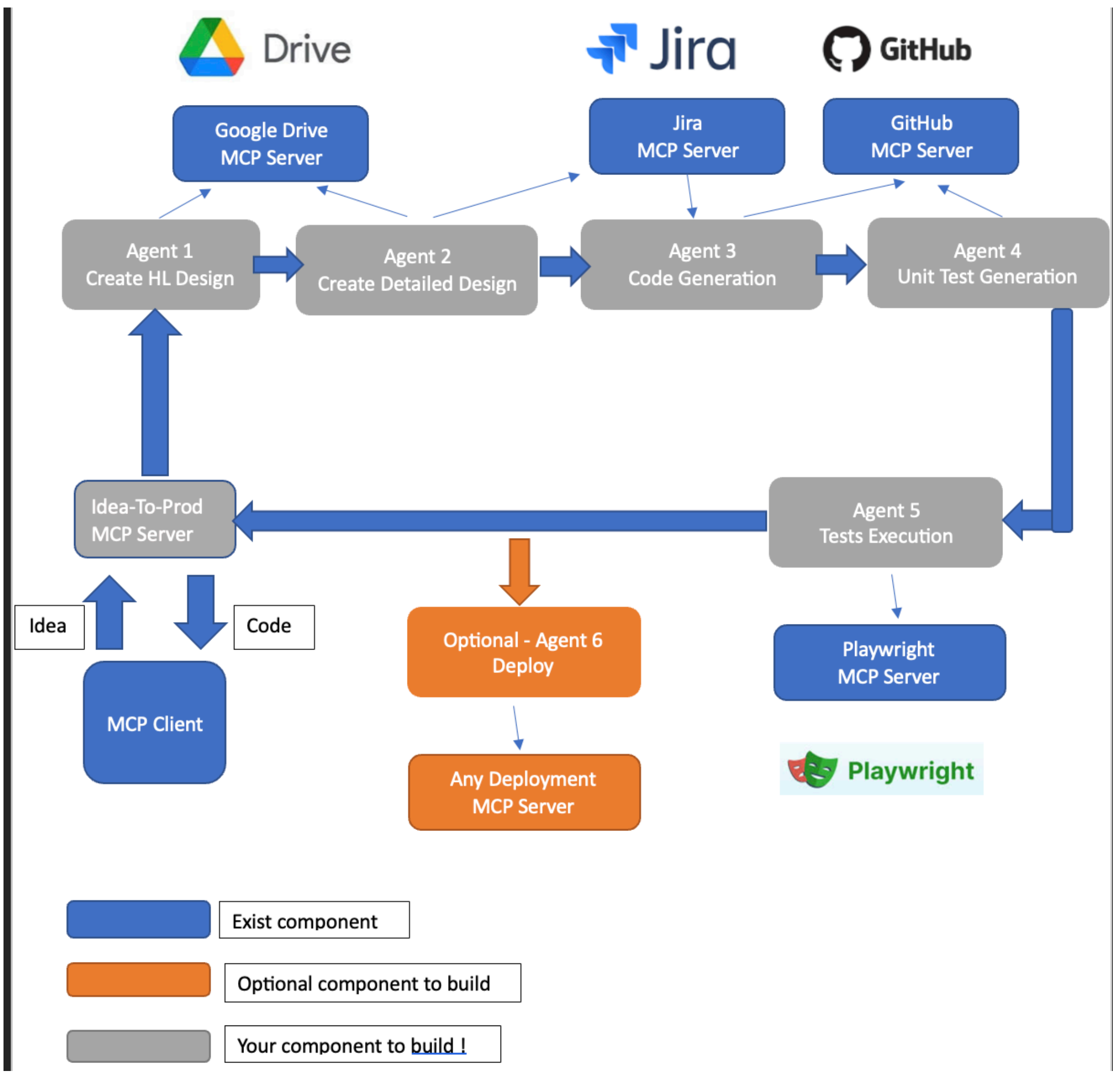
AI For Dev – Final Project

You need to build a Multi-Agent platform called Idea-To-Prod. This platform receives an input as an idea for application and generates the whole application code.

The platform build upon the following steps:

1. **The System get an idea (of building something)**
2. Creates a high level design document
3. Creates a detailed design document
4. Creates the application code and components
5. Creates unit tests for that code
6. Run the tests.
7. Optional – Deploy the application
8. **The System returns the code**

Platform Architecture



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Overview

The Multi-Agent platform is being host by a MCP Server you should build with

only one tool : **ideaToProd**. The server receives (as input) an idea for application and returns the final code. The platform composed of 5 agents + an optional one . The agents work in the following order

Agent 1 – Create HL Design

An agent that receives an idea (from a prompt) . This agent creates a high level design document with all required aspects for the application. This HL design is being saved as a document in google drive via Google Drive MCP Server

Agent 2 – Create detailed Design

An agent that receives a high level design as input, creates a very detailed design document for each phase and then does 2 things:

- Save the detailed design as a document in Google Drive via Google Drive MCP Server
- Create development tasks from the detailed design and save them as work item in Jira via Jira MCP Server

Agent 3 – Code Generation

An agent that pulled the action items from a Jira project via Jira MCP Server and generate the whole code. The code is being stored as a new GitHub repository via GitHub MCP Server.

Agent 4 – Unit Test Generation

An agent that pulled the code from the from the repository via a GitHub MCP Server and generates proper unit test. The unit tests are also been saved in GitHub.

Agent 5 – Tests Execution

An agent receives unit test cases and run them with Playwright via

Playwright MCP Server.

Only if the tests have been passed - The generated code is been returns from the Ides-To-Product MCP Server.

Optional Stages:

1. If the tests in Agent 5 are fail – The workflow "go back" to Agent 4 – for generating new code.
2. Add final stage to the process with Agent 6 that deploye the application !

Guidelines & points to consider

- Consider which models (OpenAI, Gemini, Claude, Other..) to use at each stage (Agent).
  You can check some bench marks done with those models for different usages (Which is the best model for generate documents? Which is for coding?)
- You can use any Agents tools and frameworks (LangChain / LangGraph / CrewAI / Other)
- DON'T use same models for generating the code and the generating the unit tests
- You will need to communicate with MCP Servers programmatically (from your agents). Consider using MCP-Use (https://github.com/mcp-use/mcp-use)
- Test the platform via a MCP client (Claude desktop, GitHub Copilot, other..)